

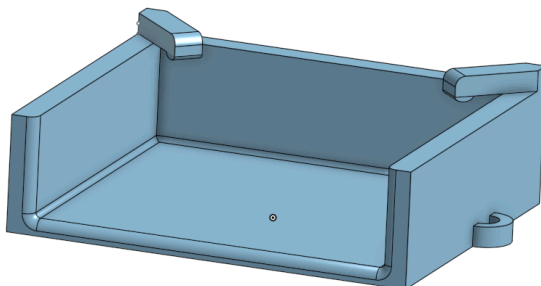
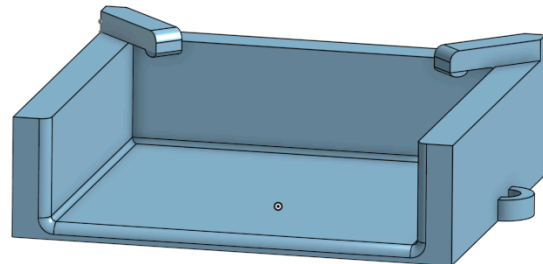
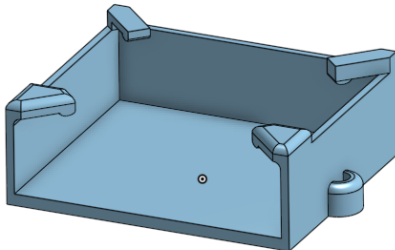
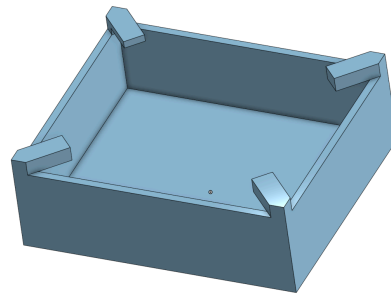
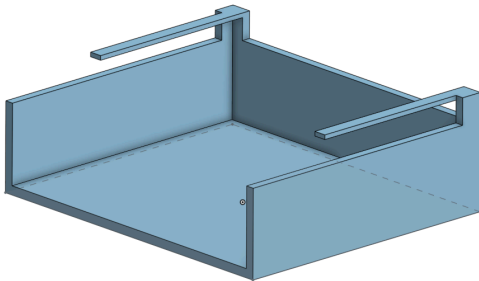
Name: Calvin, Andrew, Dylan **Date:** 4/2/2024 **Class:** Engineering Period 3

MyDesign Portfolio *Element G*

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation (G1)

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration



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Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

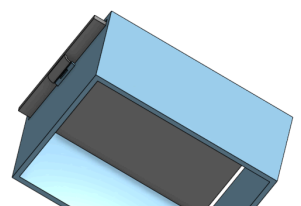
	Details	Tools and Manufacturing Methods used
Step 1	Designed CAD drawing	OnShape (2D drawing mode)
Step 2	Make revisions/corrections to CAD design	Onshape (3D modeling mode)
Step 3	Imported STL into slicing software	Prusa slider
Step 4	Print product on 3D printer	3D printer
Step 5	Attach string on side of the cover	String ordered from online
Step 6	Test product	

Final Prototype Construction of Design #2

Date: 3/12/2024

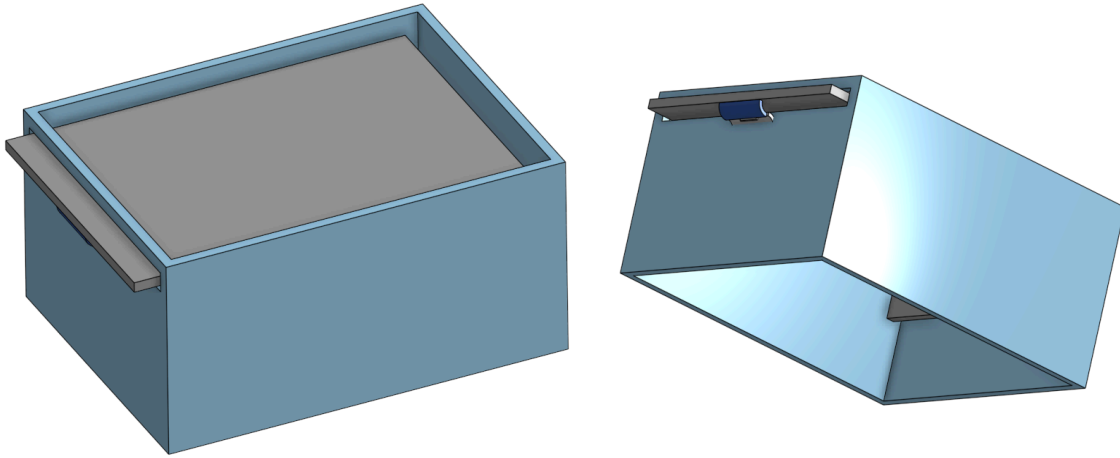
Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

This was the first edition of our design. The idea behind this design was that the door would slide back and forth, covering and protecting the camera lens when slid inwards. The door has a hook on it, which matches with the hook on the side of the cover and can grab onto it to stop the door from moving while the cover is closed. The door also



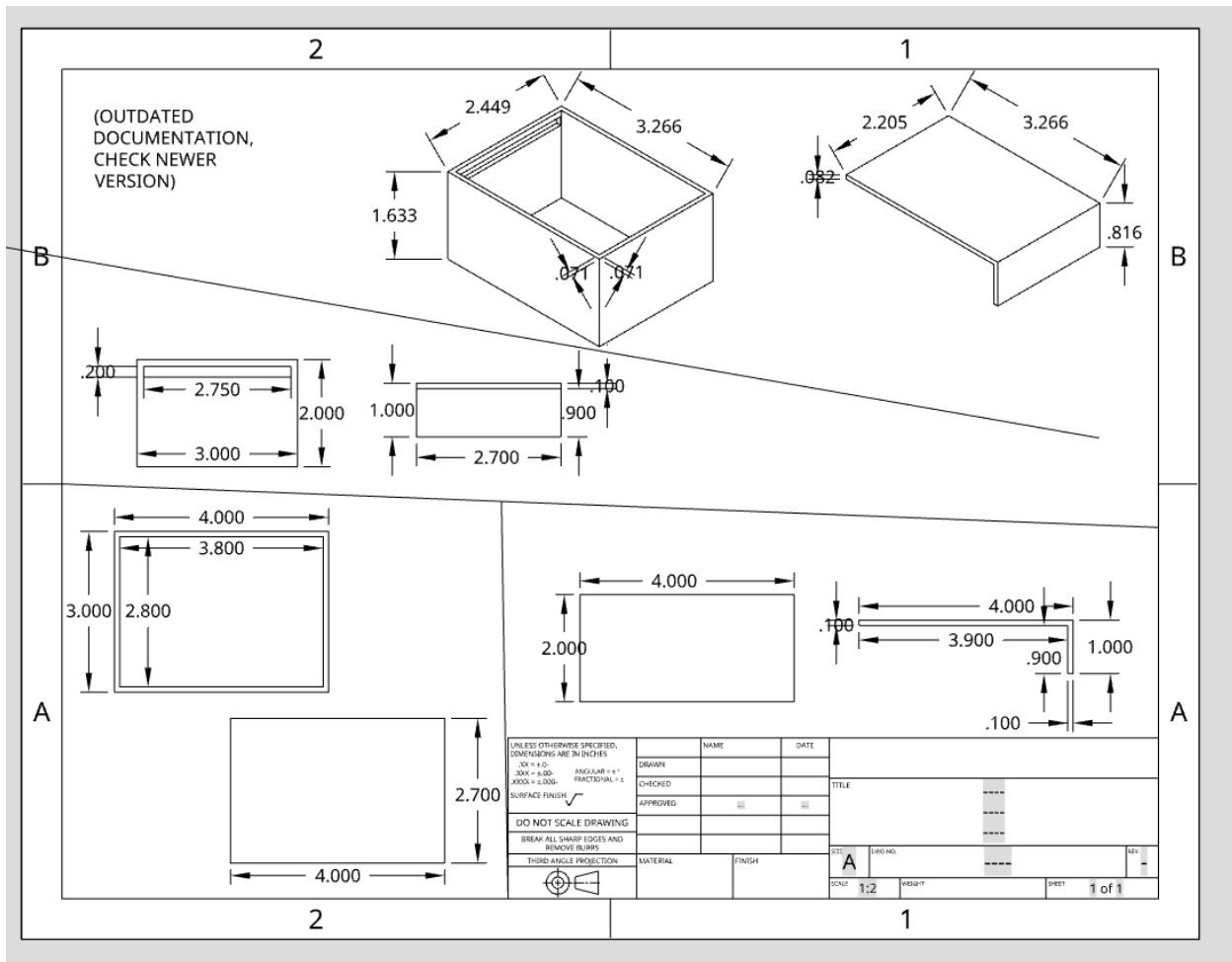
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has another piece hanging off of it, to prevent it from sliding out of the hole too easily.



Pictured: overhead view, side view, view from below.

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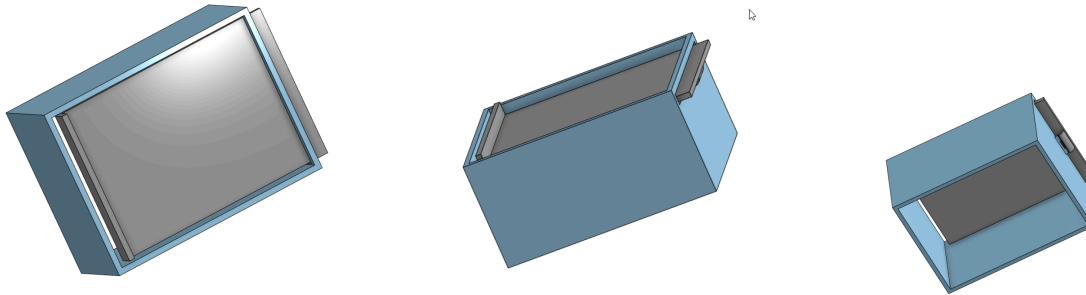


Pictured: blueprints.

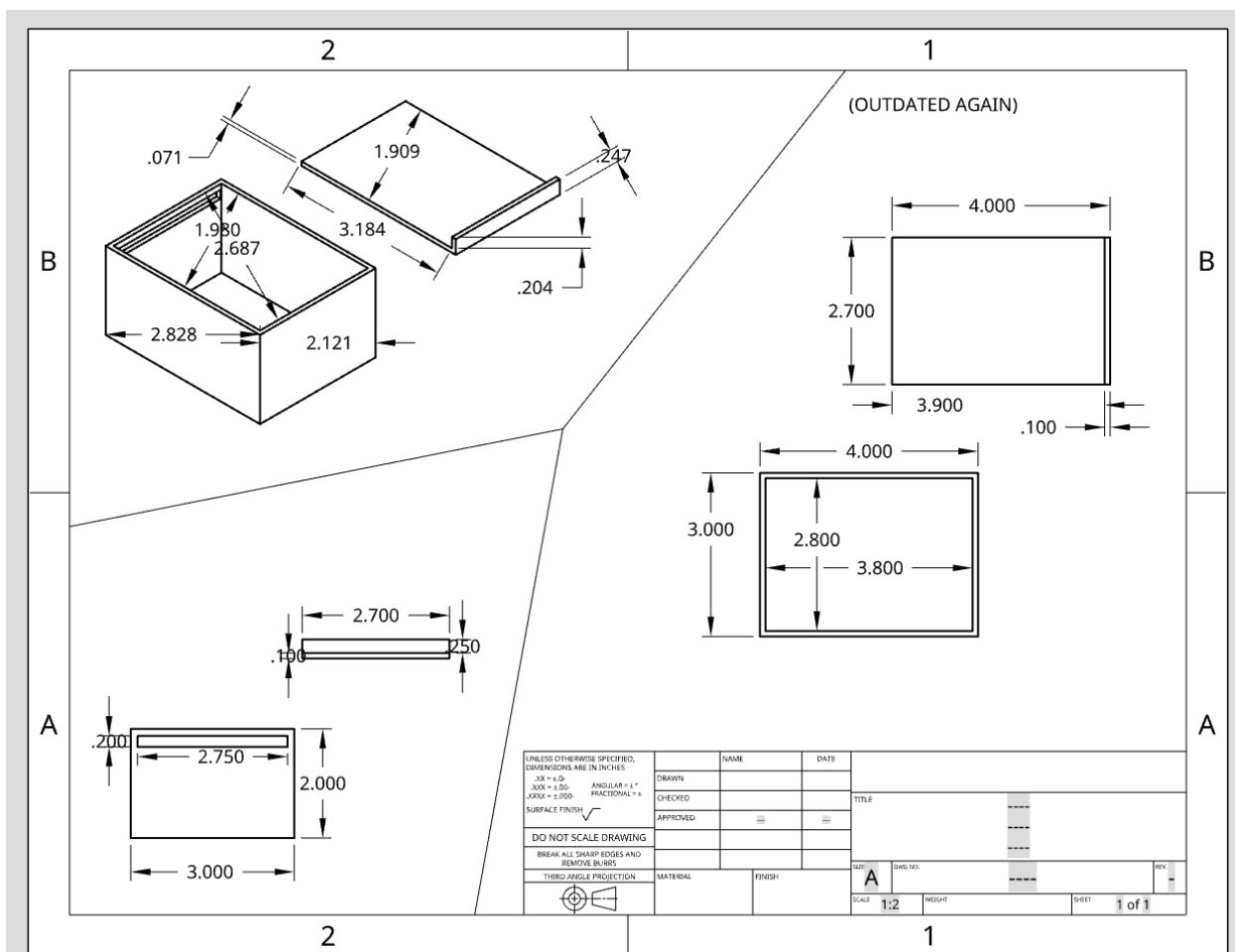
Later, we got the idea to have the piece hanging off of the door point upwards instead of downwards, to prevent it from getting in the way of the camera.



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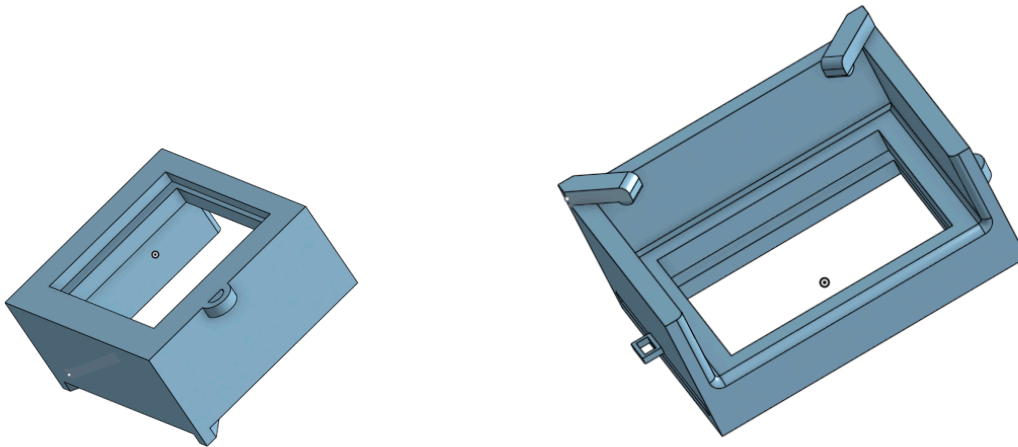
Pictured: top view, side view, bottom view.



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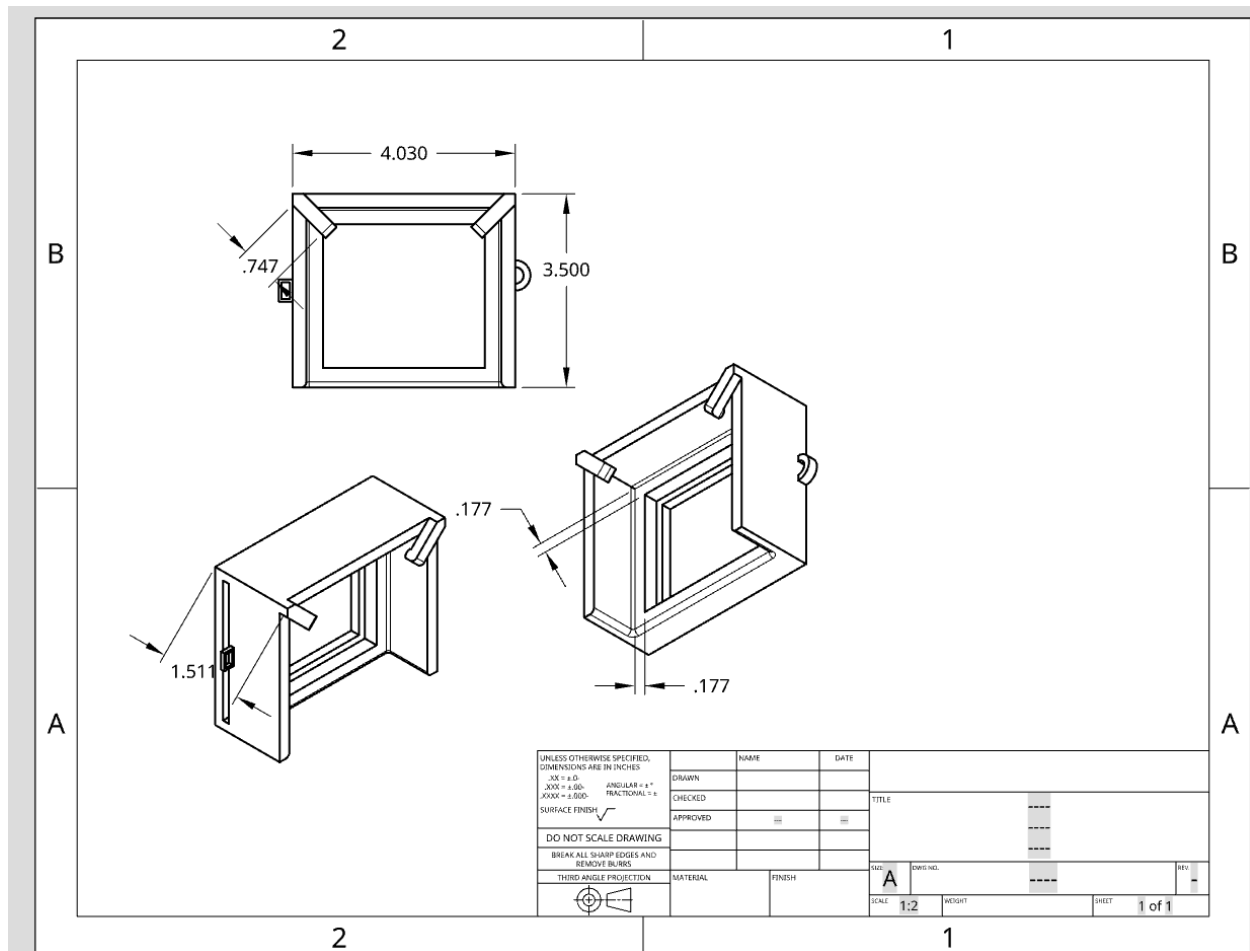
Pictured: blueprints.

I also decided to combine it with Andrew's design, which features hook-like features that can grab onto the camera lens.



Pictured: bottom view, top view.

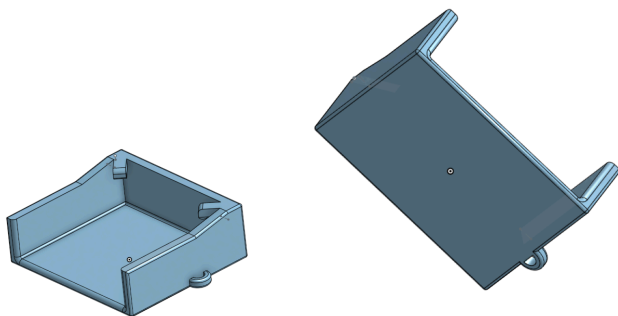
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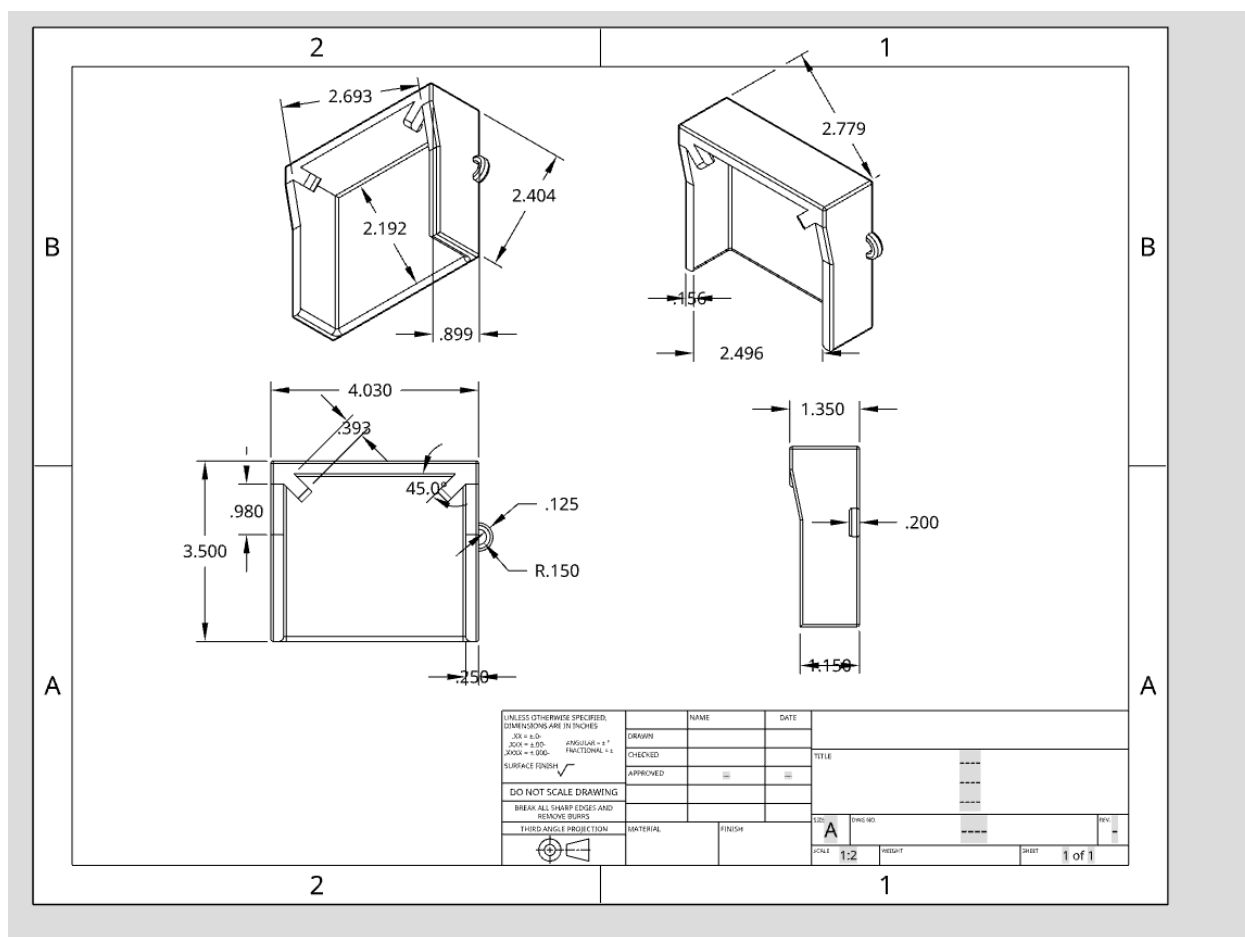
Pictured; blueprints.

Later, the idea to combine our designs was scrapped, and we decided to just go with Andrew's idea.

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Pictured: top view, bottom view.



Pictured: blueprints.

Details about all steps of construction



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	Details	Tools and Manufacturing Methods used
Step 1	Create the slide-on cover model	OnShape
Step 2	Integrate the design features of the sliding door cover into the slide-on cover	OnShape
Step 3	Export that sketch to Prusa Slicer slicing software	OnShape, Prusa Slicer
Step 4	Print out the CAD design on a 3d printer	3d printer

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

We changed the design due to failed prints and mechanical failures this caused to thicken walls, file corners, and remove unnecessary aspects

List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
3D printer	The device that will print out our design, Afinia 3D	https://afinia.com/collections/3d-printers/products/h-1-3d-printer-10x8x8-built-area-wifi-ethernet-usb-touchscreen-1-year-limited-warranty	\$2200	1	\$2200
3d print filament	The material that the	https://www.amazon.com/1-75	\$24.99 per kg	1 spool	\$2.95 on average

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	printer will use to construct the design	mm-Normal-Basic-Colors-Bundle/dp/B0BFPKZ6BR/ref=sr_1_1_sspa?c=ts&dib=eyJ2ljojMSJ9.eTkrbwU_3Eu5VbfXUdaTZLksRE4w4XXehHQ5qQ-WH3HuTUf0AnlaOSt5xRpblbGJ99Y0Q5kDBAs25P60Jw3o4Z3ZLMj-xWIAJHHI4uYrO8TQr3GwTI0UkCml9qh0GWFt0sxaN1Gqzo-NHlZUvKwBZG6vkRLx_fzL9p0EQMklBumg8twqdlE-DXCgeWhqRbfohe4ti7aFlcl7bL3deBBurID4o8gbFD3FwVWEKk-8hGjvs-lzD1kZ8CO4cbwsvG5At6L5amJil6prKrop7P5tT86SQ9hZbj0N2NH4I242Y.Du4TPUockGW_bmCS31JQGZvK6mZEoMEIMkamgqSStq4&dib_tag=se&keywords=3D%2BPrinting%2BFilament&qid=1714757719	(can produce 8 covers)	
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		https://www.thingiverse.com/thing/6066129011?from=detail&from_view=description&from_context=detail This is just one example of a filament that will work, other PLA filaments will also work			

Detailed sketches, which might include CAD if you have learned it in your class:

Construction of the Final Prototype (G2)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Protect the camera	The main goal of this design will be to protect the camera from damage, such as scratches and dust. If, during our testing, these things do get into the camera, the design will fail this criteria.
Stay attached to the lens	If this design can stay on the lens without falling off, that makes it a success here.

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Don't fall off of the camera	If the design is able to stay on the camera without getting lost, that means it succeeds at this criteria
Can be removed and reattached easily	The design will be attached and removed from the camera over and over again. If we have too much difficulty doing so, it will fail the test.
Durability	If the design is able to stay in good shape throughout the time it is used, then it is able to protect the camera and passes this criteria.
Black color	The design will be printed with black materials, meaning if it is printed with something that is too bright, it fails this criteria.
Weigh less than 50 grams	This design must be lightweight enough to not weigh down the camera. If it does weigh down the camera, that means it fails this criteria.

Testing or Modeling of Attributes (G3, G4) and Addressing Non-Testable Attributes (G4)

Attribute	Description	Describe your prototype's functionality to test this attribute
Walls	3 sides of this design have walls on them, but the 4th wall is absent to allow the camera to slide into it. The walls on the left and right sides of the design start out low, but later see a raise in elevation.	The durability tests will test how well the walls work, as if the walls survive, they will be able to protect the camera well. We will also put the cover on the camera and see if the lens is safe from damage. Lastly, the cover's weight also matters, as it shouldn't weigh down the camera too much when it is on.
Grippers	These are intended to hold the cover onto the camera while it is in use. This design has 2 grippers at diagonal angles, one at the top	Shaking the camera with the cover on will be a good test for if the grippers work, as only if the cover

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	of each walled-off corner of the design, with the claws pointing down towards the floor.	stays on will the grippers work as intended. The cover should also be easily removable and reattachable to the lens, so if the grippers grab the cover too tightly, they need to be modified.
Loop + string	The side of the design has a loop on it, on the outside of the right wall's lower edge. This exists as a place for the string to go in, so that it can be tethered to the camera.	This will be a very similar test to the grippers, except we will test it with the cover off of the lens. If the string stays within the loop, that means it is doing its job correctly. The string also needs to stay in good shape, or else the loop has nothing to tether the cover to the camera.
Floor	Lastly, the floor is what protects the lens, as it reaches across the distance of all 3 walls, which is enough length to cover the whole lens. The floor is slightly thinner than the walls, as well as also being rounded like the walls.	Just like the walls, the floor is intended to protect the camera. If the floor is able to survive the durability test and protect the camera, that means it is successful.
Color	Originally, we decided to make the color of the cover completely black, to make it match with the camera. However, following expert feedback, we realized that it would be too hard to see if it got lost somewhere. Because of this, we decided to make the cover a brighter color, like orange or blue.	This time, the cover could be tested for visibility by putting it in a place in which visibility would be a problem. If the design is visible in a dark place, that means it is bright enough to pass the test.

Addressing Non-Testable Attributes (G4)

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Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
Ergonomic design	You can't really test for ergonomics, it is more based on how it looks, which is why we can't really do physical testing.